

CHAPTER 5 REVIEW

When the vessel breaks

Pathophysiology of non-traumatic intracerebral
haemorrhage

Oxford Textbook of Stroke and Cerebrovascular Disease

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WHY IT MATTERS

The deadliest type of stroke, and outcomes have not improved



10–20%

of all strokes are intracerebral haemorrhages



40%

die within one month (median case fatality)



12–39%

regain independence in the long term

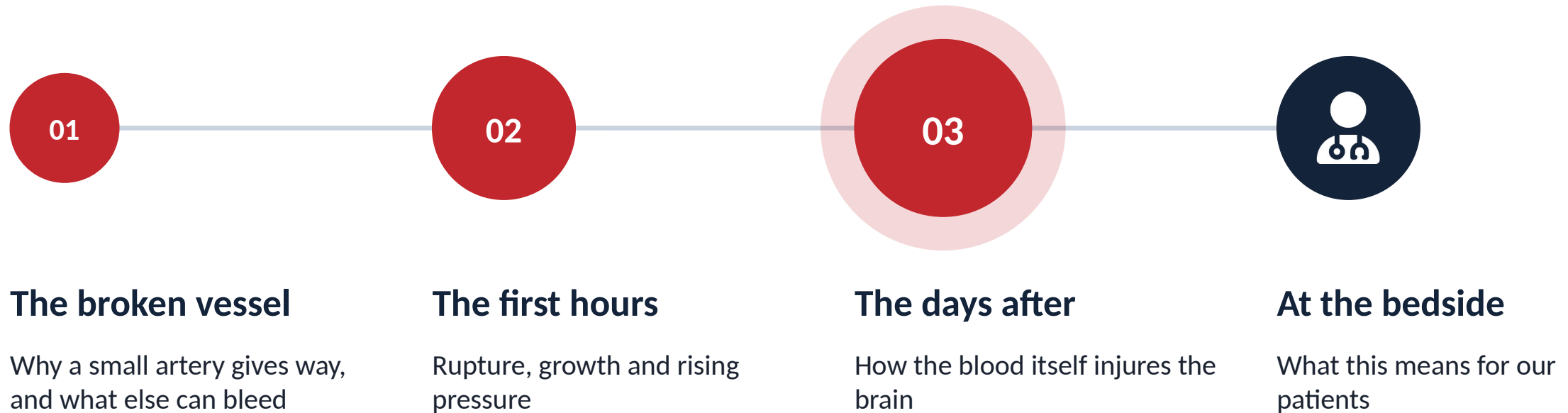
Case fatality did not fall between 1980 and 2008, **while care for ischaemic stroke moved forward.**

Sources: van Asch et al., *Lancet Neurol* 2010 (36 population-based studies, 8145 patients); Macellari et al., *Stroke* 2014.

THE BIG IDEA

ICH is a process, not a moment

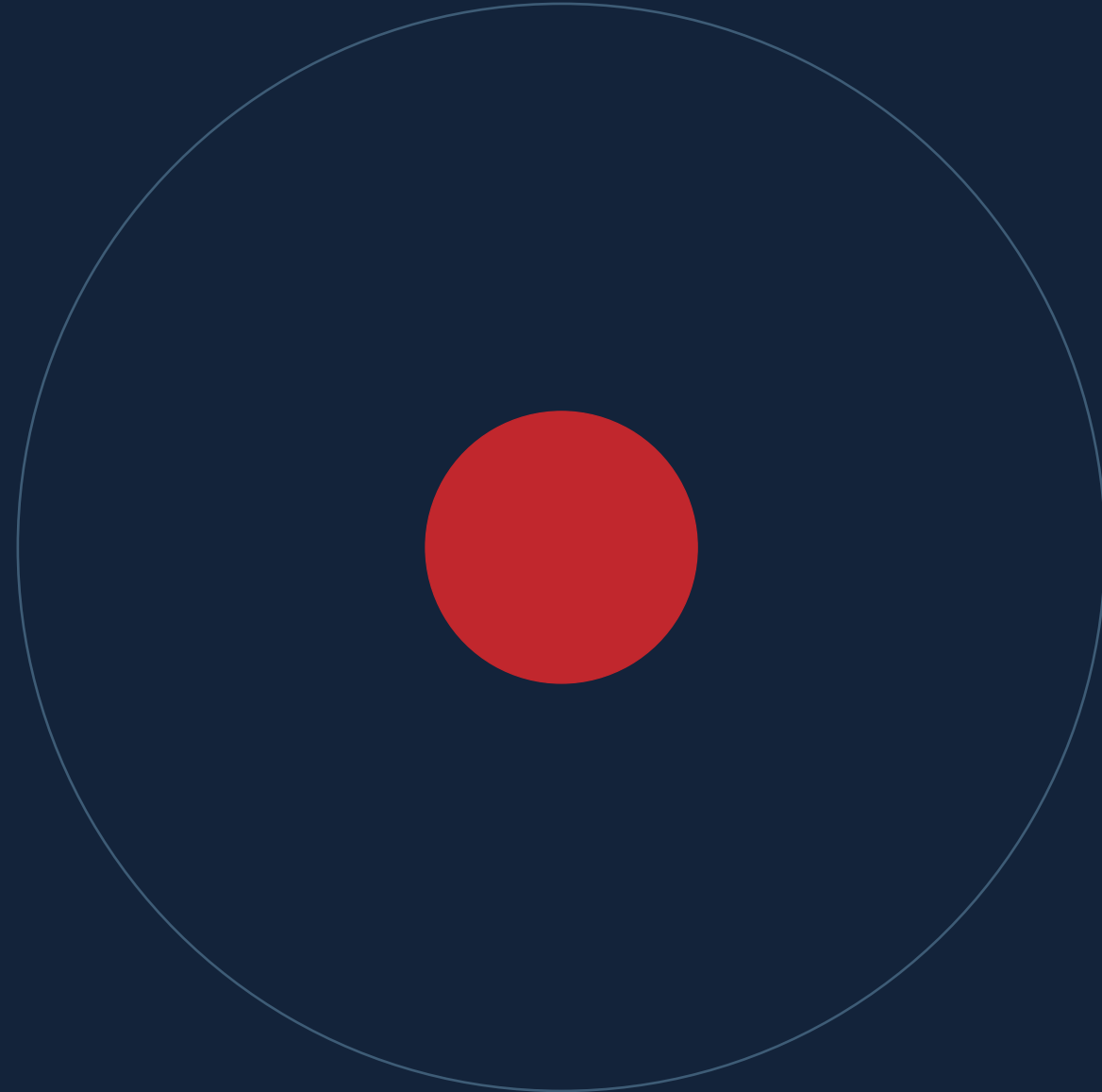
Three mechanisms unfold on three timescales, and each one gives us a chance to act.



01

The broken vessel

Why does a small artery give way, and what else can bleed?



Non-traumatic ICH has two faces: primary and secondary



Definition: bleeding directly into brain tissue without trauma. Blood may spread into the ventricles or subarachnoid space.

≈80–85%

Primary ICH

A small vessel weakened by chronic disease ruptures

1

Hypertensive arteriopathy
deep perforating arteries

2

Cerebral amyloid angiopathy
cortical and leptomeningeal arteries

≈15–20%

Secondary ICH

An identifiable lesion or condition causes the bleed

Vascular malformations

Aneurysms

Tumours

Anticoagulants

Coagulopathy

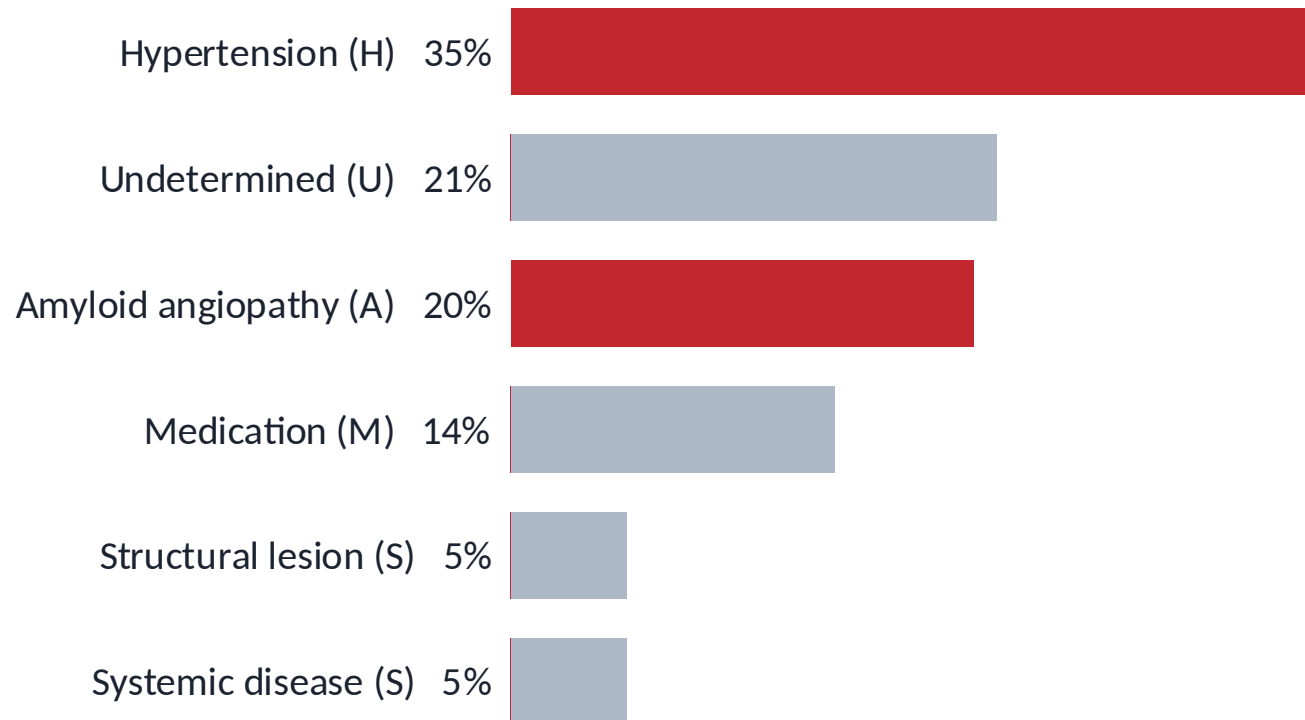
Venous thrombosis

Haemorrhagic transformation

Cocaine & amphetamines

Proportions: Macellari et al., Stroke 2014.

Two small-vessel diseases cause over half of all ICH



55%

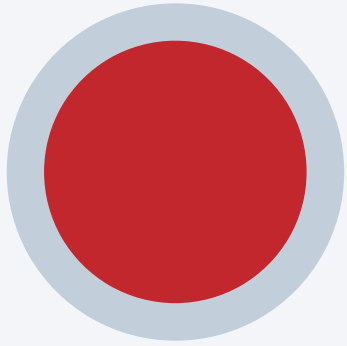
hypertensive or amyloid angiopathy:
the two small-vessel diseases

54%

of anticoagulant-related bleeds were
fatal by 3 months, the worst of any
group

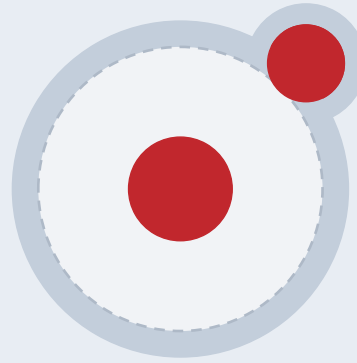
SMASH-U classification: Meretoja et al., Stroke 2012 (n = 1013, Helsinki). S = structural, M = medication, A = amyloid, S = systemic, H = hypertension, U = undetermined.

Two diseases weaken two different sets of vessels



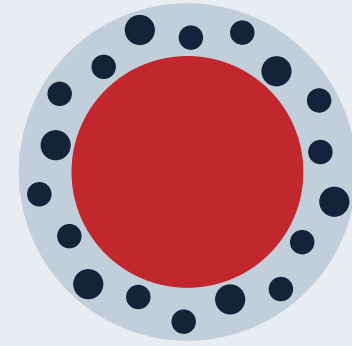
Healthy small artery

- Thin, elastic wall
- Wide lumen
- Smooth muscle buffers the pressure



Hypertensive arteriopathy

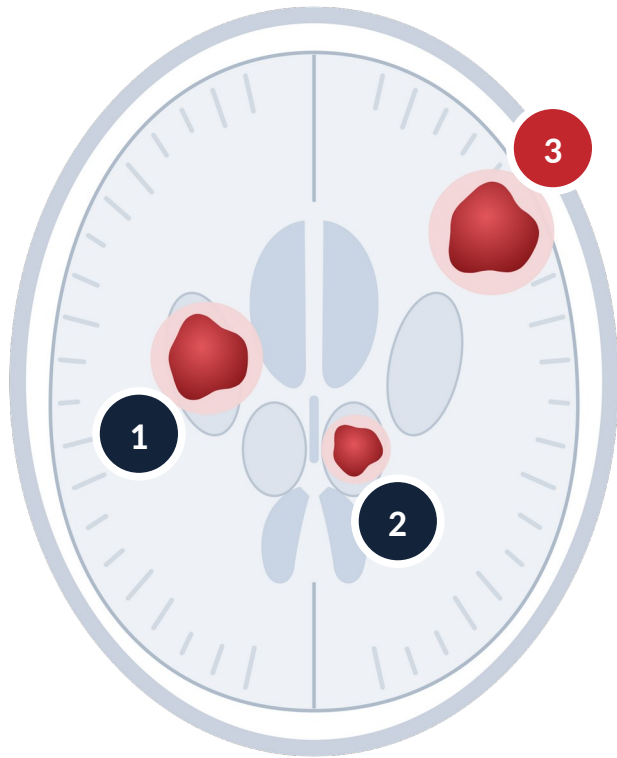
- Deep perforating arteries
- Lipohyalinosis and fibrinoid necrosis: a thick, brittle wall
- Charcot-Bouchard microaneurysms (role debated)



Cerebral amyloid angiopathy

- Cortical and leptomeningeal arteries
- β -amyloid (dark dots) replaces smooth muscle
- Mostly after 55; causes lobar bleeds

Location is the first clue to the cause



Schematic axial section; the posterior fossa (pons, cerebellum) is not shown.

1 2 Deep

Hypertensive arteriopathy

Putamen (1), thalamus (2), pons and cerebellum.

Perforating arteries leave large trunks at right angles, so they take the full force of arterial pressure.

3 Lobar

Amyloid angiopathy (older adults)

Cortex and the white matter just beneath it.

Younger, normotensive or atypical? Look for an AVM, a tumour or venous thrombosis.

Microbleeds map the disease behind the bleed



Strictly lobar
points to amyloid angiopathy

Seen in **5%** of healthy adults, **34%** after ischaemic stroke and **60%** after ICH.



Deep
points to hypertensive disease

Modified Boston criteria: diagnosing CAA in life

Definite

Full post-mortem shows severe CAA with a lobar bleed.

Probable

Age ≥ 55 ; multiple lobar or cortical bleeds, or one plus cortical superficial siderosis; no other cause.

Possible

Age ≥ 55 ; a single lobar bleed or siderosis; no other cause.



APOE $\epsilon 2$ and $\epsilon 4$ raise the risk of amyloid-related bleeding.

Cordonnier et al., Brain 2007 (systematic review); Knudsen et al., Neurology 2001; Linn et al., Neurology 2010.

Always ask: is something else causing this bleed?



Vascular malformations

AVM, cavernoma, dural fistula or aneurysm. Commonest in the young.



Antithrombotic drugs

Anticoagulants make bleeds larger and keep them growing for longer.



Tumours

Primary or metastatic. Fragile new tumour vessels bleed easily.



Venous thrombosis

Blocked drainage raises venous pressure until the tissue bleeds.



Haemorrhagic transformation

An infarct turns into a bleed, sometimes after reperfusion.



Other causes

Cocaine, amphetamines, vasculitis, RCVS, moyamoya, coagulopathy.

Some features demand a hunt for a hidden lesion



Younger patient (45 or under)



No history of hypertension



Lobar bleed, or blood only in the ventricles



CT clues: subarachnoid blood, calcification, abnormal vessels



Bleed next to a venous sinus, or a dense sinus



On anticoagulants: check clotting straight away

65%

had a vascular lesion on angiography: aged 45 or under, no hypertension, lobar bleed

0%

had one: over 45, hypertensive, with a putaminal, thalamic or posterior fossa bleed

Zhu et al., Stroke 1997 (prospective; 206 patients, all had catheter angiography).

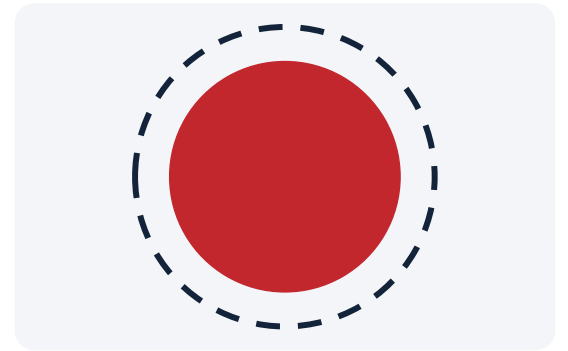
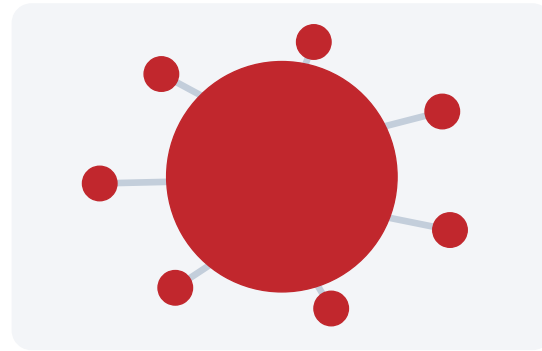
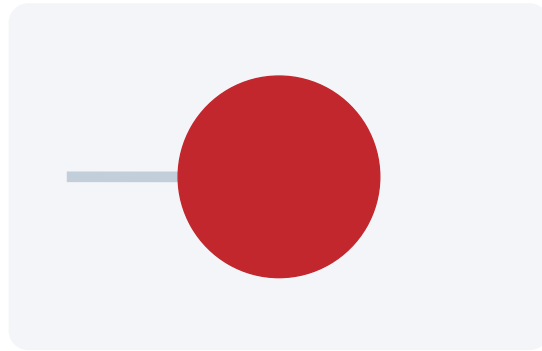
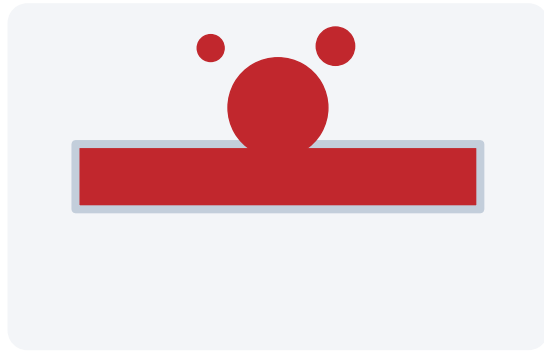
02

The first hours

Rupture, growth and rising pressure



One rupture can trigger more ruptures around it



1 Rupture

A diseased small artery gives way under arterial pressure.

2 Dissection

Blood splits along tissue planes and a haematoma forms.

3 Avalanche

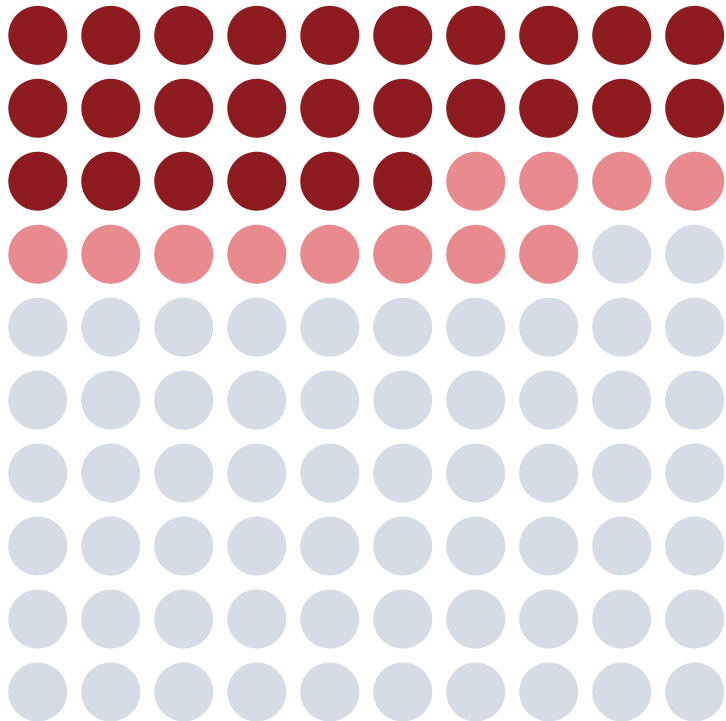
The growing clot stretches and tears neighbouring vessels at its edge (Fisher, 1971).

4 Tamponade

Bleeding stops when tissue pressure and clotting finally win.

Fisher CM, J Neuropathol Exp Neurol 1971: secondary bleeding points at the edge of hypertensive haematomas.

More than a third of haematomas keep growing



Each dot = 1 patient scanned within 3 hours of onset

38 in 100

grew by more than a third within 20 hours, and growth went with neurological decline.

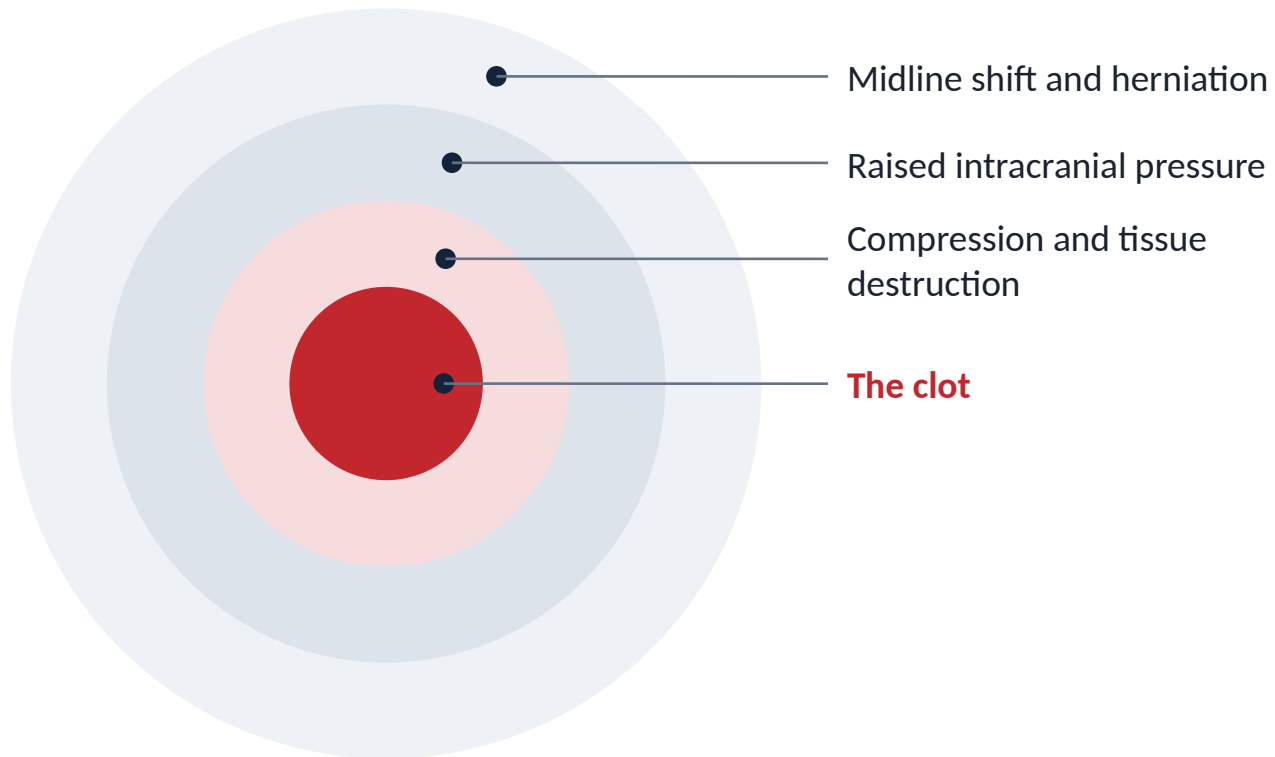
- 26 grew within the first hour after the first scan
- 12 more grew between 1 and 20 hours
- 62 showed no substantial growth



Every 10% of growth raised the hazard of death by 5%.
Higher risk: early scan, large bleed, anticoagulants, CTA "spot sign".

Brott et al., Stroke 1997 (n = 103); Davis et al., Neurology 2006; spot sign: Demchuk et al., Lancet Neurol 2012 (PREDICT).

Volume and ventricles turn a bleed into a crisis



91%

predicted 30-day mortality with volume ≥ 60 mL and GCS ≤ 8 . Volume is the strongest single predictor.

$\approx 45\%$

extend into the ventricles. Ventricular blood can cause hydrocephalus and predicts a worse outcome.

Broderick et al., Stroke 1993 (Cincinnati, n = 188); Hanley, Stroke 2009.

03

The days after

When the blood itself becomes a toxin



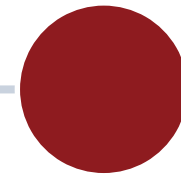
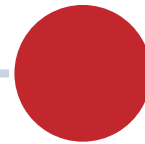
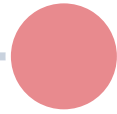
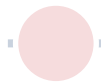
Oedema builds in waves over the first two weeks

FIRST HOURS

DAYS 1-2

DAY 3 ONWARD

WEEKS



Clot retraction

The clot contracts and squeezes serum into the surrounding tissue.

Thrombin

The clotting cascade opens the blood-brain barrier.

Red-cell lysis

Haemoglobin, haem and iron drive oxidative injury.

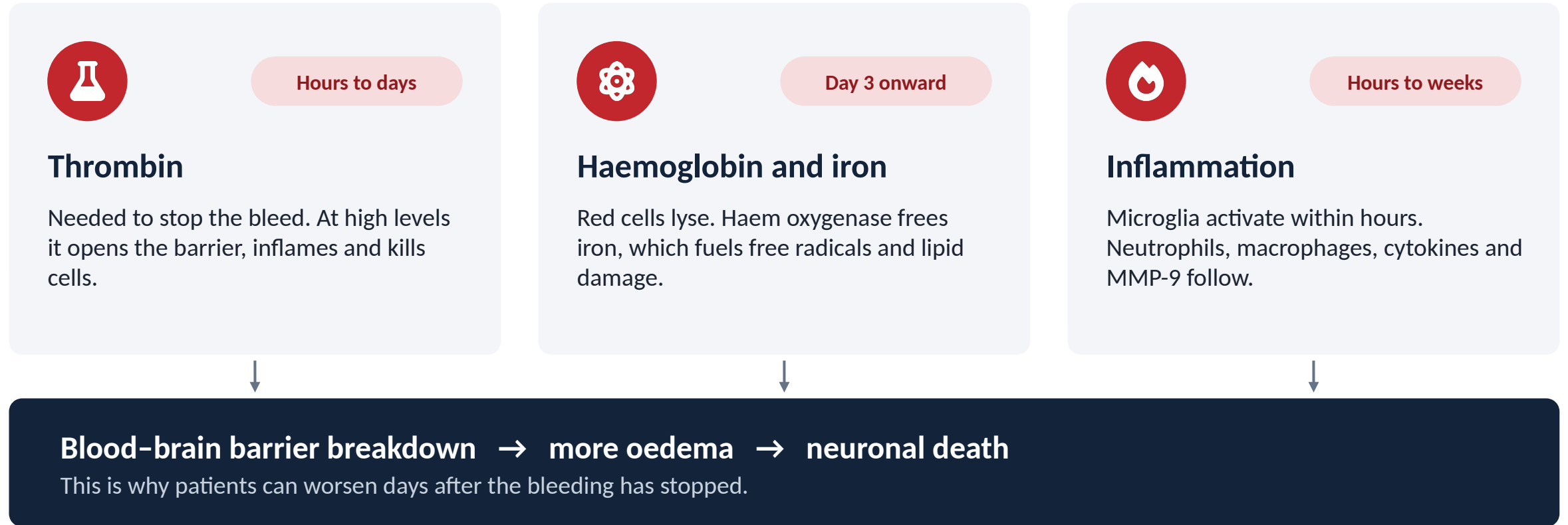
Clearance

Macrophages remove the clot, leaving a cavity and hemosiderin.

Oedema grew by about 75% in the first 24 hours and can keep building into the second week.

Xi, Keep & Hoff, Lancet Neurol 2006; oedema growth: Gebel et al., Stroke 2002.

Three toxic pathways drive secondary injury



Xi, Keep & Hoff, Lancet Neurol 2006. Most mechanistic evidence comes from animal models.

The rim around the clot is not ischaemic



The old idea

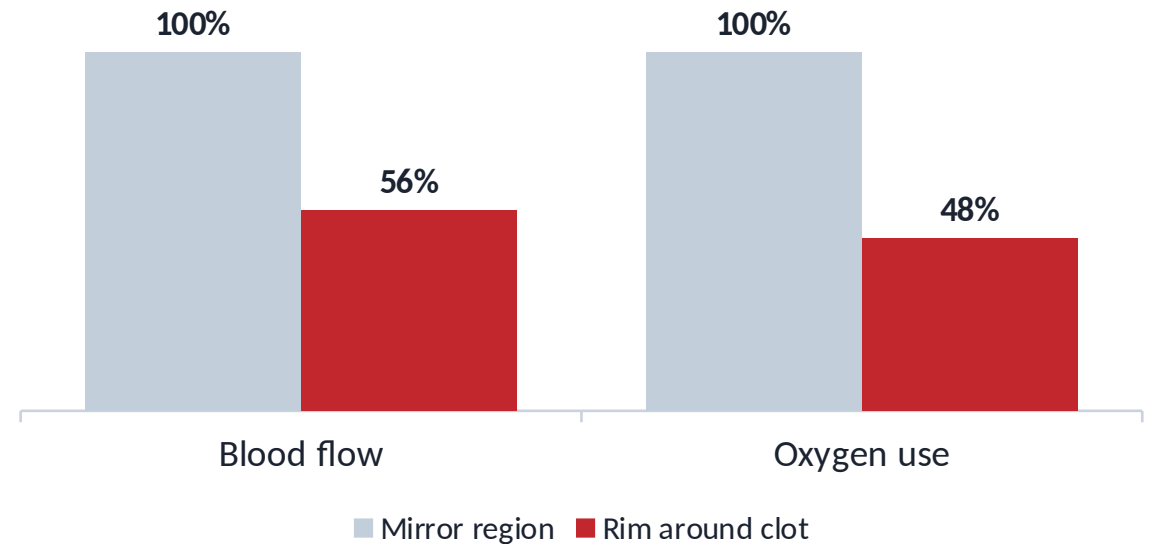
The rim is ischaemic tissue starved of blood, so lowering blood pressure could harm it.



What PET showed

Blood flow falls, but oxygen use falls further. Oxygen extraction drops: hypoperfusion without ischaemia.

Rim around the clot, as % of the mirror region



So what? Lowering systolic pressure below 150 mmHg did not reduce blood flow around the haematoma (ICH ADAPT).

Zazulia et al., *J Cereb Blood Flow Metab* 2001 (PET, n = 19, 5–22 h after onset); Butcher et al., *Stroke* 2013.

04

At the bedside

What the mechanisms mean for our patients



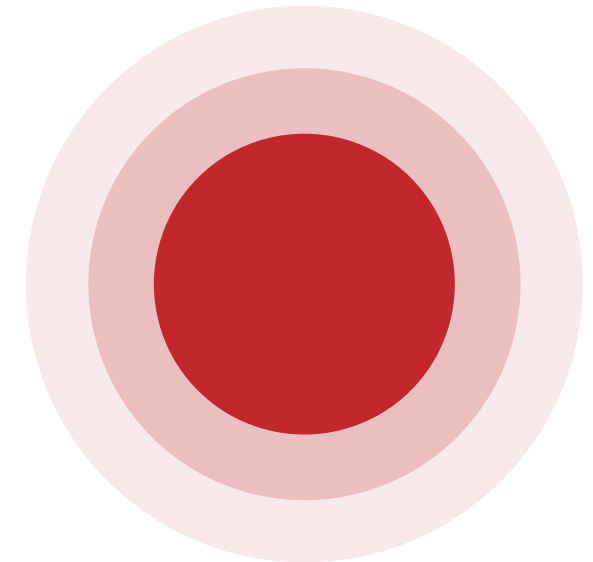
Every mechanism opens a window to act

MECHANISM	WHEN	WHAT WE DO
 Small-vessel disease	Before and after	Control blood pressure. Review antithrombotics. Older adult with a lobar bleed? Think CAA.
 Haematoma growth	First hours	Fast imaging, early blood pressure lowering and urgent anticoagulant reversal. Close neuro observations.
 Mass effect and ventricular blood	First days	New drowsiness, headache, vomiting or deficit? Stop and escalate: think growth or hydrocephalus.
 Secondary injury	Days to weeks	Expect fluctuation as oedema peaks. Reassess before each session and pace therapy.

SUMMARY

Five things to take home

- 1 **ICH is a process, not a moment.**
- 2 Deep bleeds point to hypertension. Lobar bleeds in older adults point to CAA.
- 3 Young, normotensive or atypical? Hunt for a secondary cause.
- 4 About one in three haematomas grows in the first hours.
- 5 The blood is toxic: thrombin, iron and inflammation injure the brain for days.



Vessel → hours → days

THANK YOU

Questions and discussion

When the vessel breaks, the clock starts: hours for growth, days for toxicity.



Which of these windows do we meet most often in our own practice, and are we timing our assessments around it?



SOURCES

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